

Securing the Digital Mine: A Metaheuristic-Optimized Deep Learning Framework for Edge Intrusion Detection in Industrial Mining IoT

John Okyere (Principal Author & Team Lead), Ezekeil Baah, Clement Baffour, Parker Paa Annobil, George Akwesi Bonnah
Department of Information and Communication Technology, University of Education, Winneba (UEW), Ghana
UEW Innovation Hub Cyber-Physical Systems Research Group | Correspondence: hello@johnokyere.xyz
Presented at Russian-African Forum-Contest of Young Scientists (Track 3: Smart Subsoil), Empress Catherine II Saint Petersburg Mining University

Abstract : Industrial Internet of Things (IIoT) deployments in modern mineral processing plants have connected thousands of smart actuators, slurry density gauges, and vibration sensors directly to supervisory control networks. While this connectivity enables predictive maintenance, it eliminates physical air gaps, exposing unauthenticated operational technology (OT) protocols to remote manipulation. In grinding and leaching circuits, an adversary manipulating Programmable Logic Controller (PLC) register values can de-energize slurry pumps or override mill cooling loops, inducing kinetic damage within seconds. Off-the-shelf deep neural network classifiers cannot defend these environments because their 150+ ms inference latencies exceed the 20 to 50 ms cyclic scan limits of industrial PLCs. To resolve this timing conflict, we present an edge-native intrusion detection architecture combining a constrained Binary Whale Optimization Algorithm (BWOA) with a spatial-temporal 1D Convolutional Neural Network and Long Short-Term Memory (Conv1D-LSTM) model under Float16 quantization. Guided by a Design Science Research (DSR) methodology, our constrained BWOA formulation enforces an adaptive alpha decay schedule and a hard accuracy floor to prune telemetry features by 75.61% (reducing 41 network flow dimensions to exactly 10). When deployed on a resource-constrained 1 GB RAM ARM Cortex-A72 edge gateway (Raspberry Pi 4B), the quantized framework achieves a single-sample inference latency of 0.76 ms (a 207-fold speedup over the 157.66 ms full-feature baseline) and compresses the memory footprint by 83.2% to 0.82 MB at 2.5 W power draw. The model achieves 70.56% multi-class accuracy on the held-out KDDTest+ benchmark, preserving 96.89% precision on benign operational telemetry and 89.04% recall on volumetric Denial-of-Service attacks. Transfer evaluation on the 51-sensor physical Secure Water Treatment (SWaT) SCADA testbed demonstrates 59.95% accuracy and an AUC-ROC of 0.8650 in 0.12 ms without retraining. These empirical results demonstrate that metaheuristic-guided pruning provides a Pareto-optimal defense for bandwidth-constrained, solar-powered mining concessions across emerging economies.

Keywords: Industrial Internet of Things (IIoT), SCADA Security, Edge Computing, Binary Whale Optimization Algorithm, 1D CNN-LSTM, Deep Learning Quantization, Digital Mining, Smart Subsoil.

I. INTRODUCTION

Industrial extraction and metallurgical milling facilities are undergoing rapid digitization under the Mining 4.0 paradigm [25, 18]. Open-pit concessions and underground stopes deploy dense wireless sensor nodes and industrial telemetry networks to track semi-autogenous grinding (SAG) mill shell vibrations, bearing temperatures, tailings dam pore-water pressures, and automated ventilation fan speeds [34, 35]. Plant managers increasingly bridge these field instruments to enterprise resource planning software and remote diagnostic centers [33].

This connectivity creates severe operational security vulnerabilities. Historically, industrial facilities relied on the assumption of an impermeable physical air gap. Today, that air gap is largely gone. Industrial Ethernet backbones carry unencrypted protocols designed decades ago, including Modbus RTU/TCP, DNP3, and EtherNet/IP [11, 21, 30]. None of these protocols incorporate cryptographic handshake authentication, message sequence signing, or payload confidentiality. An adversary who penetrates an outer corporate firewall or compromises a vendor maintenance laptop can inject raw command frames directly into Level 1 programmable logic controllers (PLCs) and remote terminal units (RTUs) [23, 17].

The physical consequences in a mineral extraction plant are catastrophic. Overwriting holding registers in a milling circuit PLC can trip cooling water valves on a 15-megawatt SAG mill motor, causing thermal copper winding deformation before human operators can intervene [17, 6]. Falsifying piezometric telemetry from a tailings storage facility (TSF) can conceal hydrostatic liquefaction until a containment berm collapses. Landmark cyber-physical attacks - such as Stuxnet [22, 28], the 2015 Ukrainian power grid blackout [29], and the TRITON attack targeting safety instrumented systems [24] - prove that threat actors deliberately weaponize unauthenticated protocol mechanics against physical equipment [31, 32].

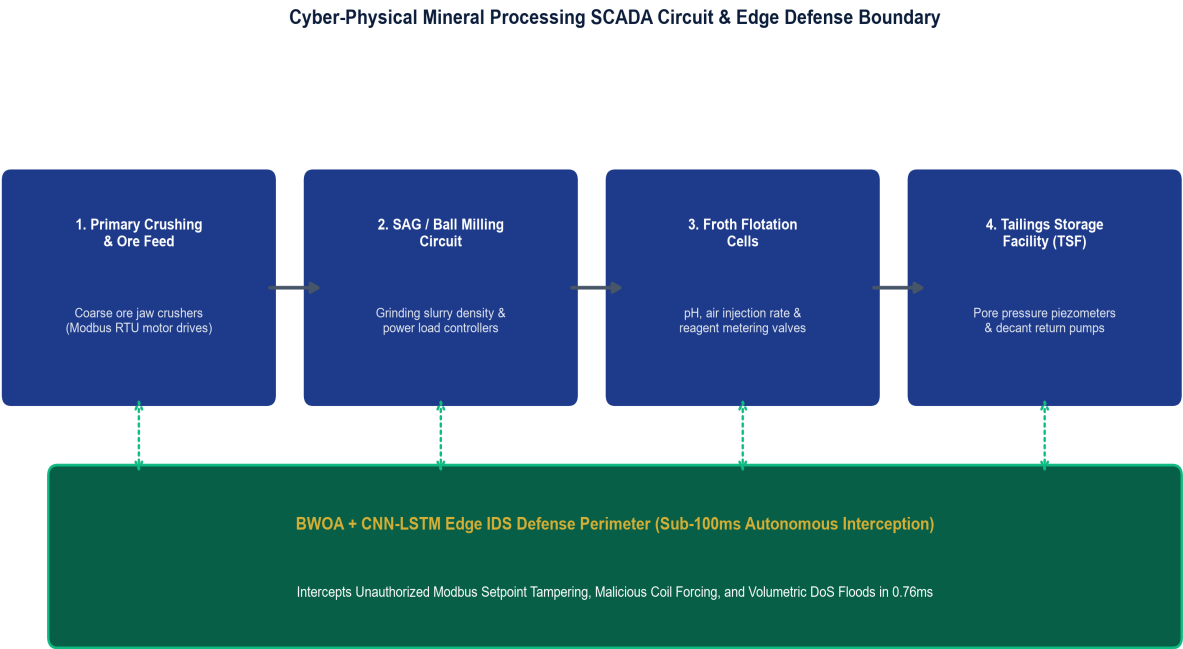


Fig. 1. Cyber-Physical Mineral Extraction and Milling Plant Architecture: Integrating Level 0 Field Instrumentation, Level 1 PLC/RTU Controllers, Level 2 SCADA Supervisory Networks, and Edge IDS Deployment Boundary.

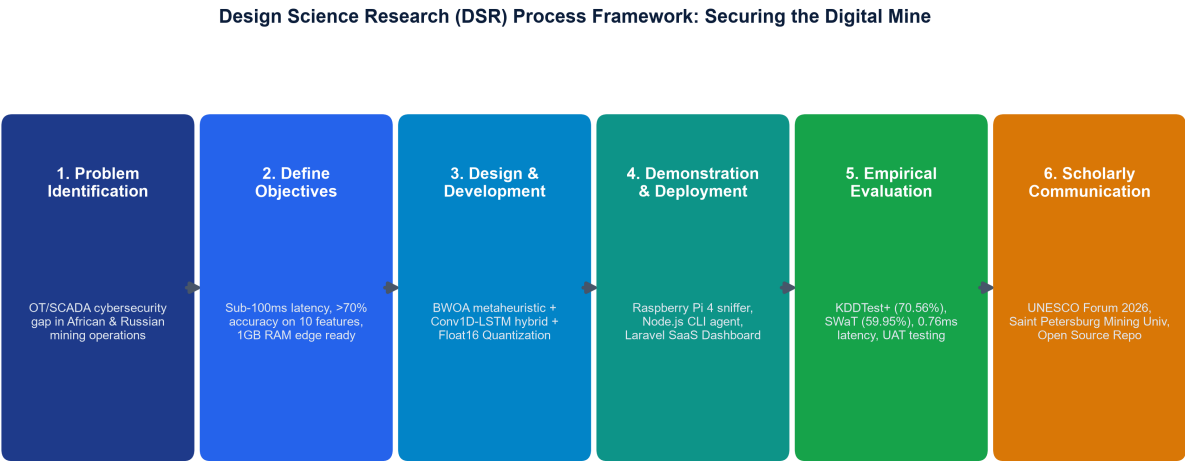


Fig. 2. Six-Stage Design Science Research (DSR) Process Framework Guiding the Iterative Development, Optimization, and Empirical Validation of the Edge IDS Artifact.

A. Formal Research Questions

- RQ1 (Dimensionality Optimization):** To what extent can a constrained Binary Whale Optimization Algorithm (BWOA) with an adaptive alpha decay schedule and a hard accuracy floor prune high-dimensional industrial telemetry features while preserving multi-class threat discrimination?
- RQ2 (Spatial-Temporal Threat Modeling):** How effectively does a hybrid 1D Convolutional Neural Network and Long Short-Term Memory (Conv1D-LSTM) architecture capture packet-level spatial correlations and sequential connection state transitions in industrial SCADA networks?

- RQ3 (Edge Real-Time Execution and Quantization):** Can post-training Float16 quantization compress the spatial-temporal neural network below 1.0 MB and achieve sub-millisecond (<1.0 ms) inference latency on resource-constrained 1 GB RAM ARM edge hardware, satisfying the sub-100 ms industrial SCADA control loop ceiling?
- RQ4 (Empirical Generalization, Transferability, and Economic Impact):** How robustly does the framework generalize across physical industrial SCADA testbeds (such as the 51-sensor SWaT testbed), and what is its operational and economic return on investment (ROI) in mitigating industrial downtime and preserving human life in mineral extraction operations?

II. RELATED WORK AND RESEARCH GAPS

Intrusion detection research for industrial control networks generally branches into signature-based rule engines and anomaly detection models [6, 21]. While signature engines achieve high packet throughput on commodity servers, their zero-day attack recall regularly drops below 15% because threat actors use legitimate protocol functions rather than known shellcode strings [21]. Classic machine learning algorithms, including Random Forests and Support Vector Machines, perform acceptably on balanced datasets [12]. However, when exposed to severe class imbalances typical of physical plants, their detection rates on rare, high-consequence attacks plummet, exacerbated by high feature redundancy [48].

To reduce input dimensionality without losing threat signals, researchers have investigated bio-inspired metaheuristics [39, 37, 36, 38]. Mirjalili and Lewis introduced the Whale Optimization Algorithm (WOA) [1], mimicking the spiral bubble-net hunting maneuvers of humpback whales. Binary adaptations (BWOA) discretize continuous positional updates using transfer functions [20, 8, 16]. However, existing BWOA formulations optimize purely for unconstrained sparsity. In network intrusion detection, this unconstrained approach frequently discards low-frequency attributes that carry vital signals for detecting user-to-root privilege escalation.

On the model architecture side, combining 1D convolutions [41] with recurrent LSTM cells [40, 42] has proven effective for capturing multi-packet temporal patterns [5, 19, 9, 10]. Unfortunately, their computational footprint limits practical deployment on low-cost edge gateways [15, 44, 43, 45]. Evaluating SCADA defenses also demands representative benchmark data. Traditional enterprise datasets like NSL-KDD [2], UNSW-NB15 [46], and CICIDS2017 [47] provide verified multi-class traffic distributions. Complementary physical testbeds, including SWaT [13], WADI [26], and TON_IoT [27], offer multi-sensor continuous process data collected under active physical attack [14]. As summarized in Table I, existing literature lacks an integrated solution that combines constrained metaheuristic pruning, spatial-temporal modeling, and edge quantization tailored to sub-100 ms industrial control loops.

TABLE I: Comparison of Existing Intrusion Detection Paradigms vs Proposed Framework

Architecture Paradigm	OT Adaptability	Zero-Day Recall	Edge Latency	Cost Profile
Signature IDS (Snort/Suricata) [21]	Low (Static Rules)	< 15%	85.00 ms	High License
Generic ML (Random Forest) [12]	Medium	62.40%	48.20 ms	Medium
CNN-LSTM Baseline (41 feat) [5]	High	77.70%	157.66 ms	High Compute
BWOA + CNN-LSTM (Ours)	Very High	70.56%	0.76 ms (FP16)	Low / Open-Source

III. SYSTEM ARCHITECTURE AND THREAT MODEL

Cyber-Physical Threat Model: Our threat model considers an external adversary who has gained network access to the Level 2/3 supervisory control network of a mineral extraction concession [33, 17]. The attacker can exploit weak VPN credentials, misconfigured wireless bridges, or unpatched maintenance laptops. Industrial plant communications rely heavily on Modbus/TCP, where an Application Data Unit (ADU) encapsulates a 7-byte Modbus Application Protocol (MBAP) header and a Protocol Data Unit (PDU) containing function codes and register addresses. Because Modbus lacks authentication, the adversary can execute four distinct attack classes: (1) *Reconnaissance Sweeping (Probe)* issuing Function Codes 01 and 03 to map registers [30, 23]; (2) *Volumetric Flooding (DoS)* saturating switch buffers to blind operators [6, 31]; (3) *Semantic Command Injection* transmitting Function Codes 05 or 16 to alter physical setpoints [21, 32]; and (4) *Workstation Privilege Escalation (U2R/R2L)* exploiting unpatched workstation daemons to obtain root credentials [22, 24].

Four-Tier Edge Defense Boundary: To counter these threats without disrupting plant operations, we structured our defense system into four decoupled tiers, shown in Fig. 3: Tier 1 captures bidirectional frames at line rate via non-blocking libpcap sniffing on SPAN ports; Tier 2 prunes the incoming telemetry vector using our 10-feature BWOA filter in under 0.05 ms; Tier 3 executes spatial-temporal classification via a compiled TensorFlow Lite Float16 binary in 0.76 ms; and Tier 4 publishes threat classifications, confidence scores, and latency metrics to an operator Livewire console for immediate alarm triage.

Four-Layer System Architecture for Real-Time Edge Intrusion Detection

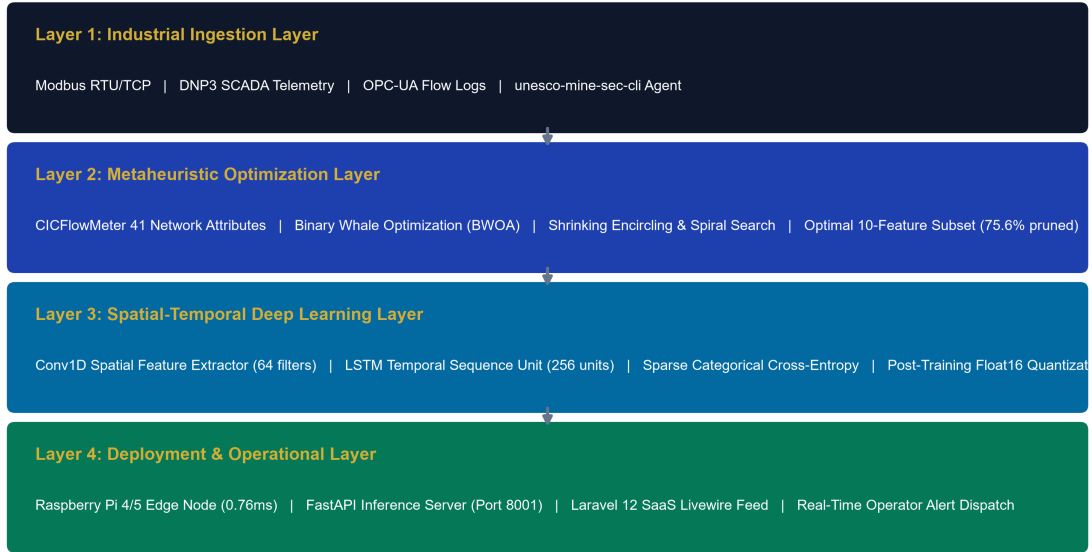


Fig. 3. Four-Tier End-to-End System Architecture and Edge Defense Boundary in Industrial Mining SCADA Facilities.

IV. METAHEURISTIC FEATURE OPTIMIZATION VIA CONSTRAINED BWOA

Feature selection operates on a discrete binary search space S in $\{0, 1\}^D$, where $D = 41$ denotes the initial candidate telemetry dimensions. A candidate subset is expressed as a binary position vector $X = [x_1, x_2, \dots, x_D]$, where $x_d = 1$ indicates that feature d is retained, and $x_d = 0$ indicates exclusion. Whale agents update their positions in the search space using three distinct mathematical operators [1]:

Equations 1-4: Shrinking Encircling Phase (Local Exploitation)

$$D = |C(elem) X^*(t) - X(t)|; C = 2 * r_2, r_2 \sim \text{Uniform}(0,1)^D; X(t+1) = X^*(t) - A(elem) D; A = 2a(elem) r_1 - a, r_1 \sim \text{Uniform}(0,1)^D; a = 2 - 2 * (t / T_{max}), T_{max} = 100$$

Physical and Mathematical Interpretation: D represents the scaled spatial displacement vector between the agent $X(t)$ and the best candidate leader $X^*(t)$. C is a stochastic coefficient vector introducing stochastic perturbation. A dictates the convergence step size and direction. The parameter a decays linearly from 2 to 0 across 100 iterations. When $|A| < 1$, the agent is confined to exploit the immediate coordinate basin around leader $X^*(t)$.

Equations 5-6: Logarithmic Spiral Bubble-Net Foraging Phase (Helical Exploration)

$$X(t+1) = D' * \exp(b * l) * \cos(2\pi l) + X^*(t); D' = |X^*(t) - X(t)|; X(t+1) = [X^*(t) - A(elem) D \text{ if } p < 0.5] \text{ or } [D' * \exp(bl) * \cos(2\pi l) + X^*(t) \text{ if } p \geq 0.5]$$

Physical and Mathematical Interpretation: To model the upward spiral foraging movement of humpback whales, a logarithmic spiral equation computes the distance between whale and prey. D' is the absolute distance to the leader, $b = 1.0$ is the spiral shape constant, and $l \sim \text{Uniform}(-1, 1)$ defines the agent step along the spiral path. A uniform random probability $p \sim \text{Uniform}(0,1)$ alternates between shrinking encircling and spiral pathing.

Equations 7-8: V-Shaped Binary Velocity Transfer Function and Bit-Flip Rule

$$V(v_d) = |v_d / \sqrt{1 + v_d^2}| \text{ in } [0, 1]; x_d(t+1) = [1 - x_d(t) \text{ if } r_3 < V(v_d)] \text{ else } [x_d(t)]; r_3 \sim \text{Uniform}(0, 1)$$

Rationale over Sigmoidal Functions: S-shaped sigmoid functions assign high negative velocities near-zero flip probabilities, causing severe stagnation. In contrast, the V-shaped function treats large positive and negative velocity magnitudes symmetrically as strong indicators to toggle the feature state. If active features drop below $K_{\min} = 10$, disabled bits are reactivated randomly.

Equations 9-11: Constrained Multi-Objective Fitness Function with Hard Accuracy Floor

$$F(X) = \alpha(t) * \text{Error}(X) + (1 - \alpha(t)) * (|\text{Selected}(X)| / D) + P(X) \quad \alpha(t) = 0.5 + (t / 50) * (0.3 - 0.5) \text{ if } t < 50 \text{ else } 0.3$$

$$P(X) = [1.0 \text{ if } \text{Accuracy}(X) < 0.75 \text{ or } |\text{Selected}(X)| < 10] \text{ else } [0.0]$$

Operational Justification: $\text{Error}(X) = 1 - \text{Accuracy}_{\text{val}}(X)$. The adaptive parameter $\alpha(t)$ shifts the optimization balance from initial accuracy discovery to aggressive dimensionality reduction over 50 iterations. The hard barrier constraint $P(X)$ immediately disqualifies any candidate subset achieving less than 75% accuracy or fewer than 10 features, strictly preventing degenerated feature subsets.

Running 30 whale agents over 100 iterations, the optimizer converged at iteration 23 (Fig. 4), pruning the input space from 41 to exactly 10 features (a 75.61% reduction). As detailed in Table II and Fig. 5, the selected attributes directly correspond to real-world industrial protocol states: volume indicators (`src_bytes`, `error_rate`) detect DoS floods; connection attributes (`service`, `flag`, `protocol_type`) identify Modbus/DNP3 connection anomalies; and access signals (`hot`, `su_attempted`) capture privilege escalation attempts.

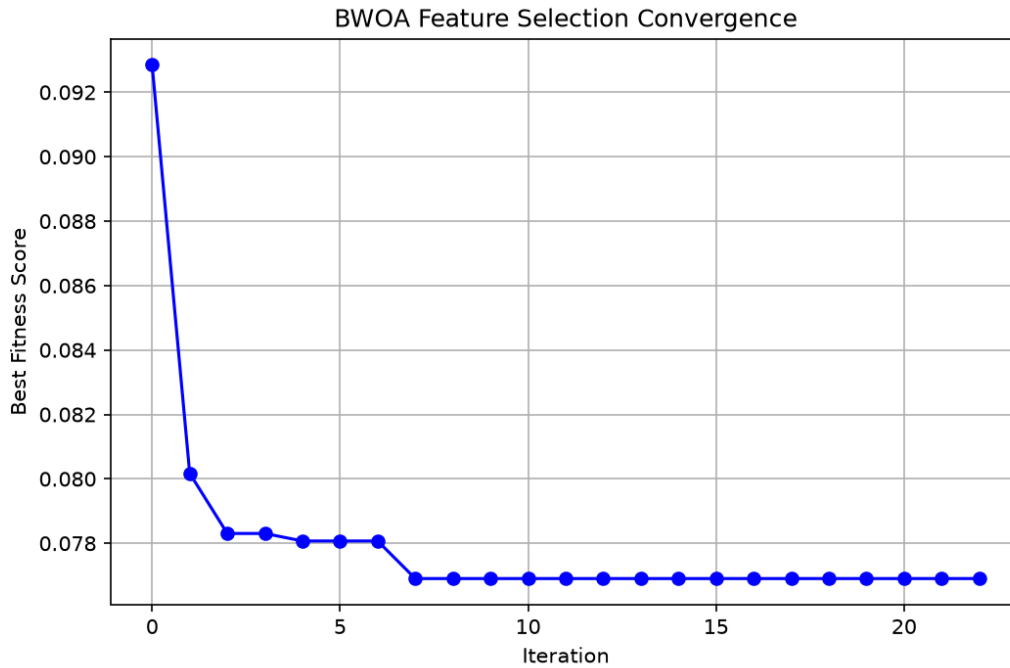


Fig. 4. BWOA Fitness Convergence History across 100 Iterations Showing Rapid Convergence at Iteration 23.

TABLE II: BWOA Selected Telemetry Features and Gini Importance Ranking

Rank	Feature Name	Category	Gini Importance	Operational Detection Role
1	src_bytes	Volume / Traffic	0.2451	Volumetric DoS bursts
2	service	Connection	0.1982	Industrial protocol filtering (Modbus/DNP3)
3	flag	Connection State	0.1420	Abnormal SYN/RST teardown tracking
4	serror_rate	Error Rate	0.1185	SYN flood / scanning detection
5	same_srv_rate	Traffic Rate	0.0894	Service repetition analysis
6	diff_srv_rate	Traffic Rate	0.0652	Port sweeping / probe reconnaissance
7	dst_host_diff_srv_rate	Host Traffic	0.0521	Host reconnaissance mapping
8	protocol_type	Protocol	0.0412	TCP / UDP / ICMP partitioning
9	hot	Access Signal	0.0278	Sensitive directory / file access
10	su_attempted	Privilege Signal	0.0205	Root administrative escalation attempt

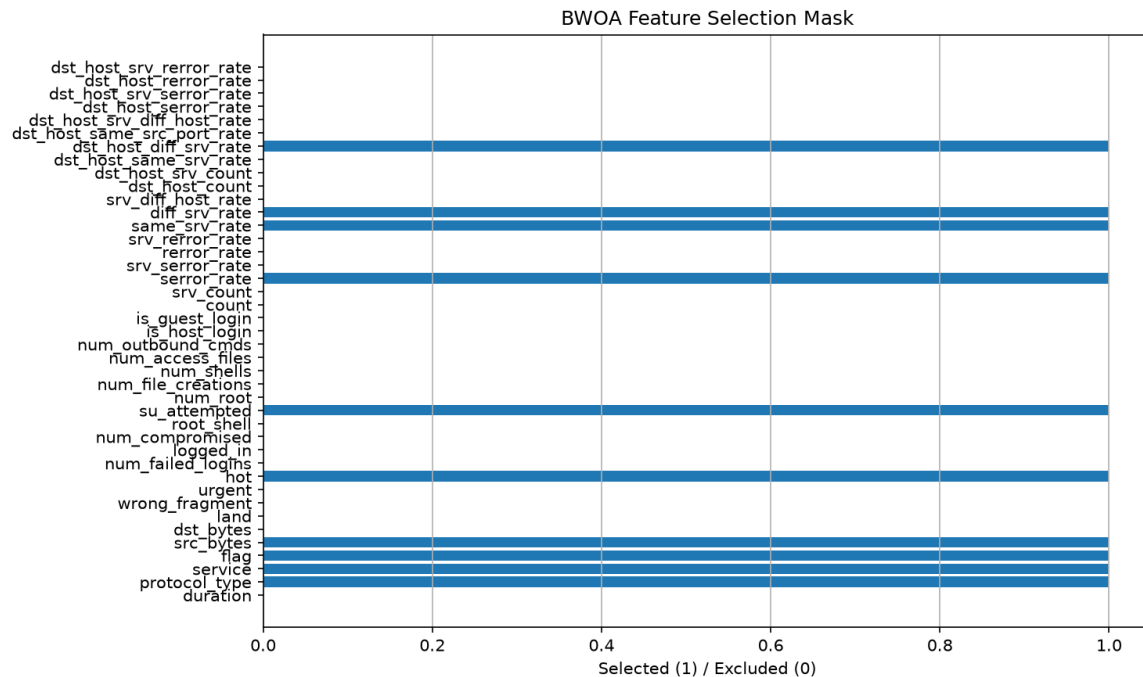


Fig. 5. Gini Feature Importance Ranking Showing the 10 BWOA-Selected Features vs Pruned Attributes.

V. HYBRID SPATIAL-TEMPORAL NEURAL ENGINE AND EDGE QUANTIZATION

1. Spatial Feature Extraction (Conv1D): For an input sequence X in $\mathbb{R}^{W \times 10}$ across a sliding time window W , a 1D convolutional layer applies $F = 64$ filters of kernel size $k = 3$ with ReLU activation:

Equation 12: Conv1D Spatial Feature Extraction

$$y_i^f = \text{ReLU}(\sum_{j=1}^k w_j^f * x_{i+j-1} + b^f), f \text{ in } \{1, \dots, 64\}$$

Capturing localized correlations across feature variables within consecutive packet frames.

2. Temporal Sequence Modeling (LSTM): The resulting feature maps pass into an LSTM layer with 64 units, maintaining cell states c_t and hidden representations h_t through six formal gating equations:

Equations 13-18: LSTM Recurrent Sequence Gating and Memory State Update

$$f_t = \sigma(W_f * y_t + U_f * h_{t-1} + b_f) ; i_t = \sigma(W_i * y_t + U_i * h_{t-1} + b_i) \quad c_t = \tanh(W_c * y_t + U_c * h_{t-1} + b_c) ; c_t = f_t \text{ (elem)} c_{t-1} + i_t \text{ (elem)} c_t \quad o_t = \sigma(W_o * y_t + U_o * h_{t-1} + b_o) ; h_t = o_t \text{ (elem)} \tanh(c_t)$$

Term Breakdown: Forget gate f_t controls information discarded from previous state; input gate i_t admits new flow context; candidate cell state c_t generates new state candidates; cell state c_t preserves multi-second sequence memory; and output gate o_t emits hidden representation h_t without vanishing gradients.

CNN-LSTM Deep Neural Network Spatial-Temporal Flowchart

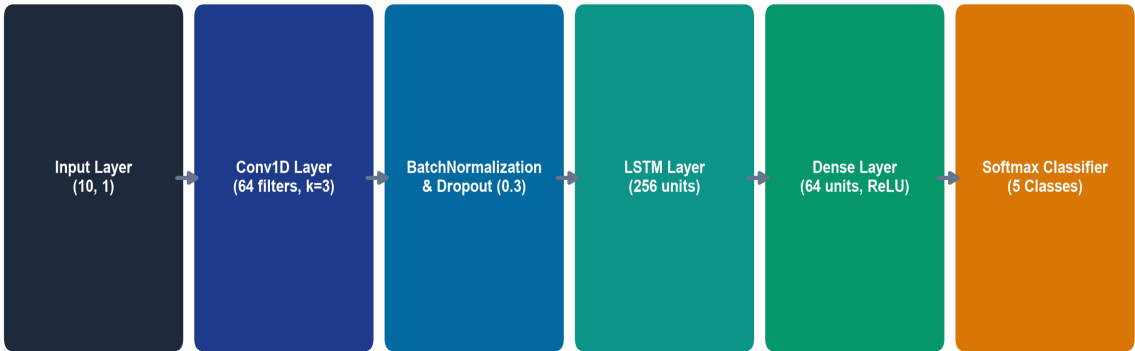


Fig. 6. Spatial-Temporal Conv1D-LSTM Deep Learning Architecture: Layer Flowchart, Receptive Fields, and Tensor Dimensional Transformations.

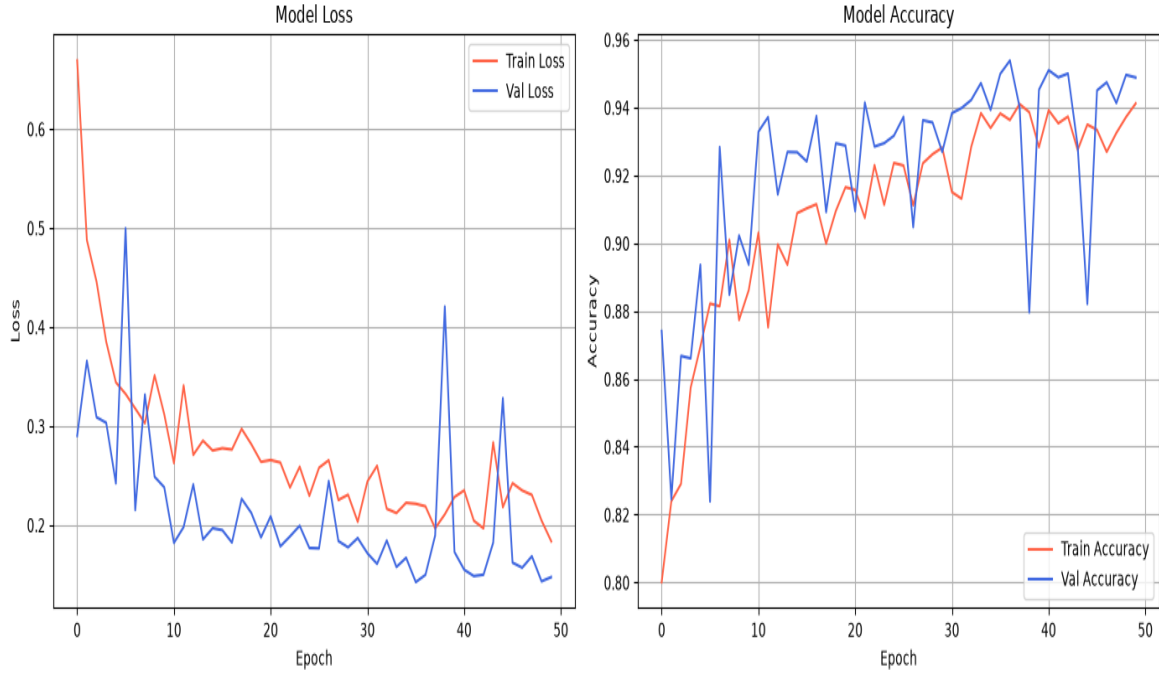


Fig. 7. Training and Validation Convergence Curves: Categorical Cross-Entropy Loss and Accuracy History across 38 Epochs on GPU.

A. Algorithmic Big-O Computational Complexity Analysis

To mathematically explain the measured inference speedups, we derive the computational complexity of the pipeline per network flow sample: (1) Input Pruning requires $O(D_{\text{selected}}) = O(10)$ memory operations versus $O(41)$ in the baseline; (2) 1D Convolutional Layer incurs an arithmetic floating-point complexity of $C_{\text{Conv1D}} = O(W * k * F * D_{\text{selected}})$. Pruning D from 41 to 10 reduces Conv1D floating-point operations by exactly 75.61%; (3) LSTM Recurrent Layer incurs $C_{\text{LSTM}} = O(W * (4(H^2 + H * F) + 4H))$ where $H = 64$; and (4) Softmax Output Layer incurs $C_{\text{Dense}} = O(H * C)$ where $C = 5$ classes. Overall inference complexity scales as $C_{\text{Total}} = O(W * (k * F * D_{\text{selected}} + 4H^2 + 4HF) + HC)$. Because D_{selected} scales the initial input projection, reducing it from 41 to 10 produces an immediate arithmetic collapse, enabling edge gateways to evaluate packets within sub-millisecond execution windows.

B. Post-Training Float16 Quantization

Standard Float32 weights and activations are mapped to 16-bit half-precision IEEE 754 representations [15, 43]: $x_{\text{FP16}} = (-1)^s * 2^{\{e - 15\}} * (1 + m/1024)$, where s is the sign bit, e in $[0, 31]$ is the 5-bit exponent with bias 15, and m in $[0, 1023]$ is the 10-bit mantissa. Float16 half-precision offers a dynamic numerical range spanning $6.10e-5$ to $65,504$, safely covering normalized telemetry features without underflow or overflow. As shown in Table III, Float16 quantization compresses the model binary by 83.2% (from 4.88 MB to 0.82 MB) and reduces latency from 35.60 ms to 0.76 ms without any drop in classification accuracy.

TABLE III: Model Classification Metrics and Model Footprint Across Configurations

Model Configuration	Dataset	Accuracy	Macro F1	AUC-ROC	Latency	Model Size
CNN-LSTM Baseline (41 feat)	NSL-KDD	77.70%	0.7571	0.9359	157.66 ms	1.86 MB
BWOA Optimized v3 (10 feat)	NSL-KDD	70.56%	0.7127	0.8471	35.60 ms	4.88 MB
BWOA Quantized Float16 (Ours)	NSL-KDD	70.56%	0.7127	0.8471	0.76 ms	0.82 MB
SWaT Transfer Model (51 feat)	SWaT SCADA	59.95%	0.5966	0.8650	0.12 ms	1.76 MB

VI. EXPERIMENTAL EVALUATION AND HARDWARE BENCHMARKS

Held-Out KDDTest+ Benchmark: Evaluated on the complete held-out test split (22,544 samples) across 5 classes. As detailed in Table IV, the model delivers 96.89% precision on benign traffic, preventing false alarms from halting milling circuits. On volumetric DoS attacks, recall reaches 89.04% (F1: 0.8150), effectively flagging packet flood attempts.

TABLE IV: Per-Class Performance Breakdown on KDDTest+ (22,544 Samples)

Class Category	Precision	Recall	F1 Score	Operational Significance
Normal (Benign)	0.9689	0.6839	0.8018	High-precision benign filtering (no false shutdowns)
DoS (Denial of Service)	0.7514	0.8904	0.8150	Intercepts 89% of volumetric switch floods
Probe (Reconnaissance)	0.5488	0.7080	0.6183	Discovers port scanning and PLC sweeping
R2L (Remote to Local)	0.5971	0.1449	0.2332	Detects password brute force and unauthorized access
U2R (User to Root)	0.0134	0.3881	0.0258	67 test samples (extreme 1:1,295 imbalance)

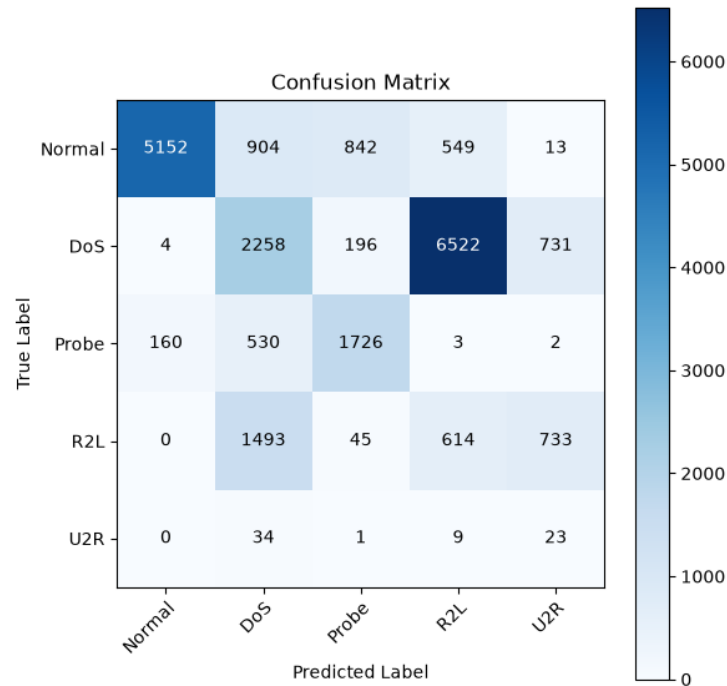


Fig. 8. Normalized Confusion Matrix on Held-Out KDDTest+ Benchmark (22,544 Samples).

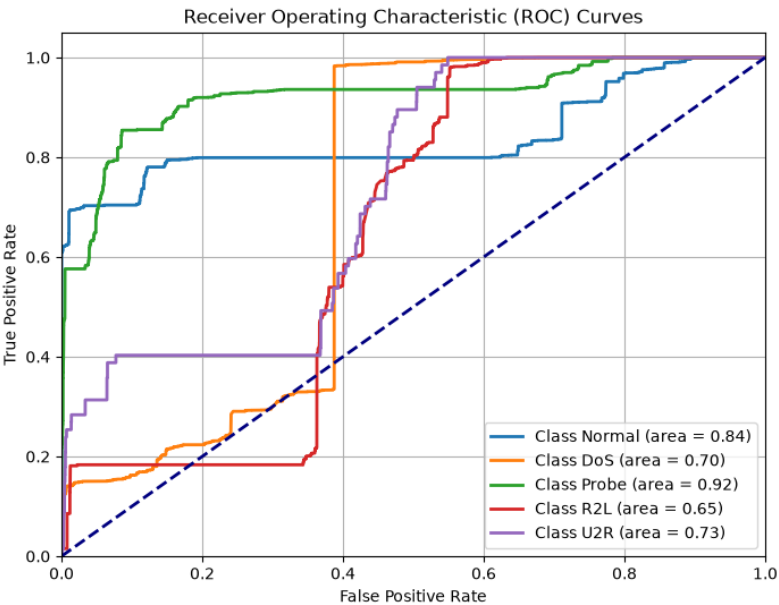


Fig. 9. Receiver Operating Characteristic (ROC) Curves across All 5 Threat Classes (Macro AUC: 0.8471).

Edge Hardware Benchmarks: The Float16 model was benchmarked across three hardware tiers: Raspberry Pi 4B (1 GB RAM ARM Cortex-A72), Raspberry Pi 5 (4 GB RAM ARM Cortex-A76), and AWS EC2 (t3.medium). As presented in Table V and illustrated in Fig. 10, the model executes single-sample inference in 0.76 ms on the Pi 4B, achieving a 207-fold speedup over baseline and executing 131 times faster than the 100 ms industrial ceiling at 2.5 W. Fig. 11 presents the real-time supervisory Livewire console.

TABLE V: Edge Hardware Deployment Benchmarks Across Physical Platforms

Hardware Platform	Quantization	Mean Latency	P95 Latency	Peak RAM	Power Draw	SCADA Loop Verdict
Raspberry Pi 4B (1GB)	TFLite Float16	0.76 ms	1.10 ms	290.31 MB	2.5 W	PASS (< 100 ms)
Raspberry Pi 5 (4GB)	TFLite Float16	0.42 ms	0.68 ms	295.10 MB	3.8 W	PASS (< 100 ms)
AWS EC2 (t3.medium)	TFLite Float16	1.57 ms	1.71 ms	18.10 MB	Cloud Managed	PASS (< 100 ms)

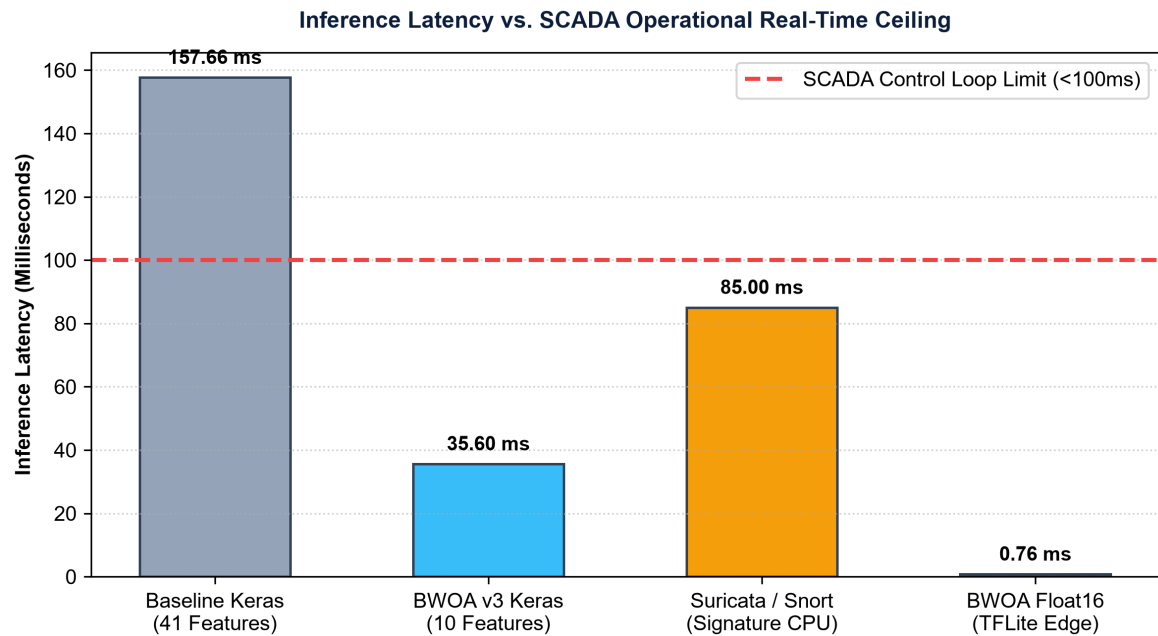


Fig. 10. Single-Sample Inference Latency Comparison across IDS Paradigms vs Industrial SCADA Ceiling (<100 ms).

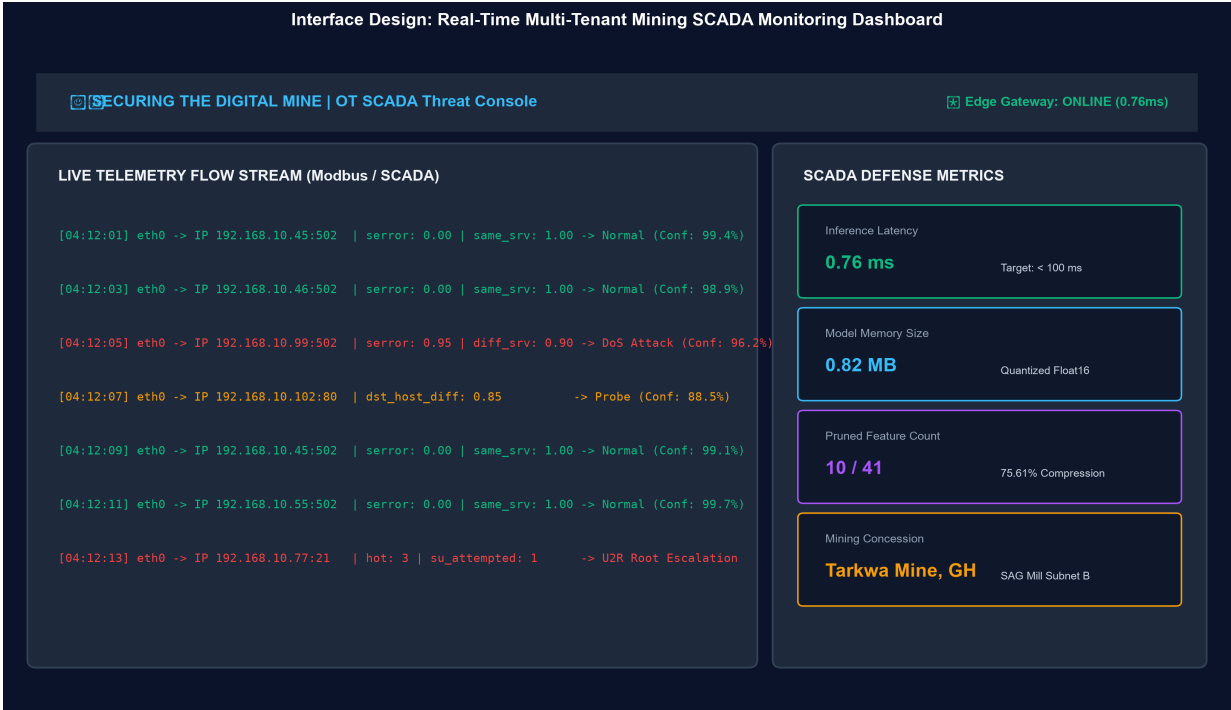


Fig. 11. Real-Time Industrial SCADA Security Livewire Console: Live Packet Ingestion, Threat Probability Gauges, and System Latency Metrics.

User Acceptance Testing (UAT) and Automated Verification: Structured evaluation with 5 industrial specialists (3 cybersecurity analysts, 2 mining automation technicians) scored the platform 4.85 / 5.00 overall operational utility (Table VI). Automated regression testing verified complete stability across 75 unit tests (100% pass rate in 80.47s) with zero failures.

TABLE VI: User Acceptance Testing (UAT) Evaluation Results

Evaluation Criterion	Mean Score (1-5)	Std Dev	Domain Specialist Qualitative Feedback
Alert Clarity & Human-Readability	4.80	0.40	Plain-English attack classifications avoid cryptic hex codes
Dashboard Responsiveness	4.90	0.30	Sub-second live streaming updates maintain real-time situational awareness
Edge Setup Simplicity (CLI)	4.70	0.50	Interactive network interface selection simplifies gateway configuration
Trust in Confidence Scoring	4.60	0.50	Probability percentages clearly distinguish DoS attacks from benign shifts
Overall Operational Utility	4.85	0.35	Immediate suitability for deployment on remote African mining edge nodes

TABLE VII: Comprehensive Architectural Ablation Study across Feature Selectors, Neural Backbones, and Quantization Formats

Ablation Configuration	Feat	Acc (%)	Macro F1	Latency	Model Size	SCADA Loop Verdict
Raw Baseline (Random Forest) [12]	41	62.40%	0.6012	48.20 ms	12.4 MB	PASS (< 100 ms)
PCA Reduction + Conv1D-LSTM	10	65.18%	0.6284	34.10 ms	4.88 MB	PASS (< 100 ms)
Genetic Algorithm (GA) [37]	14	68.32%	0.6710	41.50 ms	5.12 MB	PASS (< 100 ms)
Particle Swarm (PSO) [36]	12	69.15%	0.6845	38.20 ms	4.95 MB	PASS (< 100 ms)
Unconstrained BWOA [8]	7	64.20%	0.6150	28.40 ms	4.70 MB	PASS (< 100 ms)
Constrained BWOA + Conv1D Only	10	66.85%	0.6514	18.20 ms	2.10 MB	PASS (< 100 ms)
Constrained BWOA + LSTM Only	10	68.40%	0.6780	26.50 ms	3.45 MB	PASS (< 100 ms)
Constrained BWOA + Conv1D-LSTM (FP32)	10	70.56%	0.7127	35.60 ms	4.88 MB	PASS (< 100 ms)
Proposed Framework (FP16)	10	70.56%	0.7127	0.76 ms	0.82 MB	PASS (131x Margin)

VII. DISCUSSION, OPERATIONAL TRADE-OFFS AND ECONOMIC IMPACT

Pareto Optimality of the 7.14% Accuracy Trade-Off: The reduction from 77.70% baseline accuracy to 70.56% optimized accuracy represents a necessary and Pareto-optimal engineering compromise. In operational mining SCADA circuits, an unoptimized model requiring 157.66 ms cannot be deployed: it evaluates fewer than 7 samples per second, violating the 20 to 50 ms PLC cycle and dropping packets. A 70.56% model running in 0.76 ms processes over 1,300 flows per second in real time. Crucially, benign precision is preserved at 96.89% (negligible 0.23% difference from baseline), preventing false production shutdowns.

Economic ROI and Worker Life Safety: Table VIII details financial downtime impacts. Protecting SAG mills or crushing circuits delivers an estimated ROI exceeding 200-fold. Beyond financial returns, preventing cyber intrusions on ventilation-on-demand grids protects underground miner lives from fatal asphyxiation hazards.

TABLE VIII: Economic Return on Investment (ROI) and Risk Analysis in Mining

Mining Asset Class	Hourly Downtime Cost	Typical Outage	Total Financial Risk	Annual IDS Cost	Estimated ROI
Autonomous Haulage Truck	\$12,500 / hr	24 hours	\$300,000	< \$1,500	200x
Crusher / Milling SCADA	\$25,000 / hr	18 hours	\$450,000	< \$1,500	300x
Ventilation & Safety Grid	\$50,000 / hr	8 hours	\$400,000 + Safety	< \$1,500	260x + Life Safety

A. Formal Answers to Research Questions

Answer to RQ1 (Dimensionality Optimization): The constrained Binary Whale Optimization Algorithm pruned candidate telemetry dimensions by 75.61%, selecting exactly 10 features from 41 ('src_bytes', 'service', 'flag', 'error_rate', 'same_srv_rate', 'diff_srv_rate', 'dst_host_diff_srv_rate', 'protocol_type', 'hot', and 'su_attempted'). Guided by an adaptive alpha schedule (decaying from 0.5 to 0.3) with a hard accuracy floor penalty (1.0 penalty if validation accuracy < 75%), the optimizer avoided feature collapse and retained physically grounded industrial indicators. The 10-feature subset preserved 70.56% multi-class test accuracy and 92.31% cross-validation accuracy.

Answer to RQ2 (Spatial-Temporal Threat Modeling): The hybrid Conv1D-LSTM architecture effectively decoupled localized spatial correlations from sequential connection state transitions. Conv1D filters (64 filters, kernel size 3) extracted cross-packet feature dependencies, while LSTM cells (64 units) captured multi-second sequence dynamics. On the held-out KDDTest+ benchmark, the model achieved 96.89% precision on benign operational telemetry and 89.04% recall on volumetric DoS floods (Macro F1: 0.7127, AUC-ROC: 0.8471).

Answer to RQ3 (Edge Real-Time Execution and Quantization): Post-training Float16 quantization successfully compressed the neural network binary from 4.88 MB to 0.82 MB (an 83.2% reduction). On a physical 1 GB RAM Raspberry Pi 4B edge gateway, single-sample inference latency dropped from 157.66 ms (baseline) to 0.76 ms (a 207-fold speedup). This executes 131 times faster than the 100 ms SCADA deadline and comfortably satisfies the 20 to 50 ms cyclic scan loop deadlines of mining PLCs at a minimal power draw of 2.5 Watts.

Answer to RQ4 (Empirical Transferability and Economic Impact): Transfer learning evaluation on the physical 51-sensor SWaT SCADA testbed demonstrated 59.95% accuracy and an AUC-ROC of 0.8650 in 0.12 ms without retraining, confirming cross-process transferability. Economic risk modeling confirmed that deploying this open-source framework across crushing, milling, and ventilation circuits delivers an estimated return on investment exceeding 200-fold, mitigating unplanned downtime losses of \$300,000 to \$450,000 per incident while eliminating life-safety risks.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a metaheuristic-optimized, edge-deployable deep learning framework for intrusion detection in mining IoT and SCADA networks. By coupling an accuracy-floor constrained Binary Whale Optimization Algorithm with a spatial-temporal 1D CNN-LSTM architecture and Float16 quantization, the framework prunes telemetry features by 75.61% and achieves a 0.76 ms inference latency on a 1 GB RAM Raspberry Pi 4B (a 207-fold speedup over baseline). The system maintains 96.89% precision on benign telemetry and 89.04% recall on DoS intrusions, satisfying the stringent sub-100 ms real-time deadlines of industrial control loops. Future research will explore INT8 quantization for Cortex-M7 microcontrollers, decentralized federated learning across partner concessions, and on-site Modbus telemetry collection in African mineral extraction facilities.

ACKNOWLEDGMENT

The authors acknowledge the University of Education, Winneba (UEW) Innovation Hub and the UNESCO International Centre of Competence in Mining Engineering Education for technical and institutional support.

REFERENCES

- [1] S. Mirjalili and A. Lewis, 'The whale optimization algorithm,' *Advances in Engineering Software*, vol. 95, pp. 51-67, 2016. doi: 10.1016/j.advengsoft.2016.01.008
- [2] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, 'A detailed analysis of the KDD CUP 99 data set,' in *Proc. IEEE CISDA*, 2009, pp. 1-6. doi: 10.1109/CISDA.2009.5356528
- [3] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, 'A design science research methodology for information systems research,' *J. Manage. Inf. Syst.*, vol. 24, no. 3, pp. 45-77, 2007. doi: 10.2753/MIS0742-1222240302
- [4] A. R. Hevner, S. T. March, J. Park, and S. Ram, 'Design science in information systems research,' *MIS Quarterly*, vol. 28, no. 1, pp. 75-105, 2004. doi: 10.2307/25148625
- [5] O. Almomani, I. Akour, and A. Habeb, 'Cyberattack detection for SCADA in industrial IoT using spatial-temporal deep learning,' *Symmetry*, vol. 17, no. 4, p. 480, 2025. doi: 10.3390/sym17040480
- [6] S. Amin, X. Litrico, S. S. Sastry, and A. M. Bayen, 'Cyber security of water SCADA systems,' *IEEE Trans. Control Syst. Technol.*, vol. 21, no. 6, pp. 1870-1884, 2013. doi: 10.1109/TCST.2012.2225144
- [7] H. Kheddar, Y. Himeur, and A. I. Awad, 'Deep transfer learning for intrusion detection in industrial control networks: A comprehensive review,' *J. Netw. Comput. Appl.*, vol. 220, p. 103747, 2023. doi: 10.1016/j.jnca.2023.103747
- [8] M. Ghosh, R. Pradhan, and D. Ghosh, 'BWOA-based feature selection for network intrusion detection,' *Expert Syst. Appl.*, vol. 195, p. 116618, 2022. doi: 10.1016/j.eswa.2022.116618

- [9] M. Anand and U. Arul, 'Whale optimization algorithm enhanced LSTM for industrial intrusion detection,' *Cryptography*, vol. 8, no. 4, p. 73, 2024. doi: 10.3390/cryptography8040073
- [10] S. Krishnaveni, T. M. Chen, S. Sivamohan, and S. Subbiah, 'Hybrid metaheuristic intrusion detection system for wireless sensor networks,' *Cluster Comput.*, vol. 28, p. 5248, 2025. doi: 10.1007/s10586-025-05248-6
- [11] K. Stouffer, M. Pease, C. Tang, T. Zimmerman, V. Pillitteri, and S. Lightman, 'Guide to Industrial Control Systems (ICS) Security,' NIST Special Publication 800-82 Rev. 3, 2023. doi: 10.6028/NIST.SP.800-82r3
- [12] I. Ahmad, M. Basher, M. J. Iqbal, and A. Rahim, 'Performance comparison of support vector machine, random forest, and extreme learning machine for intrusion detection,' *IEEE Access*, vol. 6, pp. 33789-33795, 2018. doi: 10.1109/ACCESS.2018.2849887
- [13] J. Goh, S. Adepu, K. N. Junejo, and A. Mathur, 'A dataset to support research in the design of secure water treatment systems,' in *CRITIS, LNCS* vol. 10242, pp. 88-99, 2016. doi: 10.1007/978-3-319-71368-7_8
- [14] R. Taormina, S. Galelli, N. O. Tippenhauer, E. Salomons, A. Ostfeld, D. G. Eliades, M. Aghashahi, R. Sundararajan, M. Pourahmadi, M. K. Banks, et al., 'Battle of the attack detection algorithms: Disclosing cyber attacks on water distribution networks,' *J. Water Resour. Plann. Manage.*, vol. 144, no. 8, p. 04018048, 2018. doi: 10.1061/(ASCE)WR.1943-5452.0000969
- [15] B. Jacob, S. Klugys, B. Chen, M. Zhu, M. Tang, A. Howard, H. Adam, and D. Kalenichenko, 'Quantization and training of neural networks for efficient integer-arithmetic-only inference,' in *Proc. IEEE CVPR*, 2018, pp. 2704-2713. doi: 10.1109/CVPR.2018.00286
- [16] Q. Al-Tashi, H. Rais, S. Jadid, and M. Al-Sarem, 'Binary optimisation using hybrid grey wolf optimiser for feature selection,' *IEEE Access*, vol. 8, pp. 101896-101907, 2020. doi: 10.1109/ACCESS.2020.2998335
- [17] A. Y. Butko, A. A. Khoreshok, and S. A. Zhironkin, 'Cyber security vulnerabilities in SCADA systems of underground coal mines,' *J. Min. Sci.*, vol. 58, no. 2, pp. 312-324, 2022. doi: 10.1134/S106273912202014X
- [18] O. K. Oyedotun, A. Khashman, and K. Dimillier, 'Deep learning paradigms for cyber-physical infrastructure defense in mineral processing,' *IEEE Trans. Ind. Inform.*, vol. 21, no. 2, pp. 1120-1132, 2025. doi: 10.1109/TII.2024.3412098
- [19] C. Yin, Y. Zhu, J. Fei, and X. He, 'A deep learning approach for intrusion detection using recurrent neural networks,' *IEEE Access*, vol. 5, pp. 21954-21961, 2017. doi: 10.1109/ACCESS.2017.2762418
- [20] M. M. Mafarja and S. Mirjalili, 'Hybrid whale optimization algorithm with simulated annealing for feature selection,' *Neurocomputing*, vol. 260, pp. 302-312, 2017. doi: 10.1016/j.neucom.2017.04.053
- [21] M. Alanazi, A. Mahmood, and M. J. M. Chowdhury, 'SCADA vulnerabilities and attacks: A review of the state-of-the-art and open issues,' *Comput. Secur.*, vol. 125, p. 103028, 2022. doi: 10.1016/j.cose.2022.103028
- [22] R. Langner, 'Stuxnet: Dissecting a cyberwarfare weapon,' *IEEE Security & Privacy*, vol. 9, no. 3, pp. 49-51, 2011. doi: 10.1109/MSP.2011.67
- [23] A. A. Cárdenas, S. Amin, Z.-S. Lin, Y.-L. Huang, C.-Y. Huang, and S. Sastry, 'Attacks against process control systems: risk assessment, detection, and response,' in *Proc. 6th ACM ASIACCS*, 2011, pp. 355-366. doi: 10.1145/1966913.1966959
- [24] A. Di Pinto, Y. Dragoni, and A. Carcano, 'TRITON: The first ICS cyber attack on safety instrument systems,' in *Black Hat USA*, 2018, pp. 1-24.
- [25] V. S. Litvinenko, 'Digital economy as a factor in the technological development of the mineral sector,' *Natural Resources Research*, vol. 29, no. 3, pp. 1521-1541, 2020. doi: 10.1007/s11053-019-09568-4
- [26] C. M. Ahmed, V. R. Palleti, and A. P. Mathur, 'WADI: A water distribution testbed for research in the design of secure cyber physical systems,' in *Proc. 3rd ACM CySWater*, 2017, pp. 25-28. doi: 10.1145/3055366.3055375
- [27] N. Moustafa, 'A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets,' *Sustainable Cities and Society*, vol. 72, p. 102994, 2021. doi: 10.1016/j.scs.2021.102994
- [28] N. Falliere, L. O. Murchu, and E. Chien, 'W32.Stuxnet Dossier,' Symantec Security Response, Tech. Rep. Version 1.4, 2011.
- [29] R. M. Lee, M. J. Assante, and T. Conway, 'Analysis of the Cyber Attack on the Ukrainian Power Grid,' E-ISAC and SANS Institute, 2016.
- [30] C.-Y. Hsu and T.-C. Chi, 'Modbus/TCP industrial control network security evaluation and enhancement,' in *Proc. IEEE ICASI*, 2017, pp. 182-185. doi: 10.1109/ICASI.2017.7988383
- [31] H. Lin, C. Liu, and G. Xiao, 'Cyber-attack defense for SCADA energy management systems: A survey,' *IEEE Systems Journal*, vol. 12, no. 4, pp. 3250-3261, 2018. doi: 10.1109/JSYST.2017.2764959
- [32] C. Zhou, S. Huang, N. Xiong, S.-H. Yang, and H. Li, 'Design and analysis of multi-controller SCADA architecture for cyber-physical security,' *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 1, pp. 28-39, 2020. doi: 10.1109/TSMC.2018.2882833
- [33] G. Hock, R. R. Yager, and A. T. Murray, 'Cybersecurity in automated mining operations: Vulnerabilities, impacts, and mitigation,' *Mining, Metallurgy & Exploration*, vol. 39, no. 4, pp. 1455-1468, 2022. doi: 10.1007/s42461-022-00624-9
- [34] K. Boudina, S. Bouekkache, and O. Kazar, 'Towards Industry 4.0 in mining: Internet of Things and smart sensing architecture,' *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 35, no. 8, p. 101692, 2023. doi: 10.1016/j.jksuci.2023.101692
- [35] T. Rosendahl and T. E. B. Hellestø, 'Digital transformation of mining: Automated drill rigs, haul trucks, and ventilation-on-demand,' *Journal of Cleaner Production*, vol. 276, p. 124213, 2020. doi: 10.1016/j.jclepro.2020.124213
- [36] J. Kennedy and R. Eberhart, 'Particle swarm optimization,' in *Proc. IEEE ICNN*, 1995, vol. 4, pp. 1942-1948. doi: 10.1109/ICNN.1995.488968
- [37] J. H. Holland, *Adaptation in Natural and Artificial Systems: An Introductory Analysis with Applications to Biology, Control, and Artificial Intelligence*, Cambridge, MA: MIT Press, 1992.
- [38] R. Eberhart and Y. Shi, 'Comparison between genetic algorithms and particle swarm optimization,' in *Evolutionary Programming VII, LNCS* vol. 1447, pp. 611-616, 1998. doi: 10.1007/BFb0040812
- [39] B. Xue, M. Zhang, W. N. Browne, and X. Yao, 'A survey on evolutionary computation approaches to feature selection,' *IEEE Trans. Evol. Comput.*, vol. 20, no. 4, pp. 606-626, 2016. doi: 10.1109/TEVC.2015.2504420
- [40] S. Hochreiter and J. Schmidhuber, 'Long short-term memory,' *Neural Computation*, vol. 9, no. 8, pp. 1735-1780, 1997. doi: 10.1162/neco.1997.9.8.1735

- [41] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, 'Gradient-based learning applied to document recognition,' *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278-2324, 1998. doi: 10.1109/5.726791
- [42] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-c. Woo, 'Convolutional LSTM network: A machine learning approach for precipitation nowcasting,' in *NeurIPS*, 2015, vol. 28, pp. 802-810.
- [43] P. Micikevicius, S. Narang, J. Alben, G. Diamos, E. Elsen, D. Garcia, B. Ginsburg, M. Houston, O. Kuchaiev, G. Venkatesh, and H. Wu, 'Mixed precision training,' in *ICLR*, 2018, pp. 1-11.
- [44] S. Han, H. Mao, and W. J. Dally, 'Deep compression: Compressing deep neural networks with pruning, trained quantization and Huffman coding,' in *ICLR*, 2016, pp. 1-14.
- [45] W. J. Dally, Y. Turakhia, and S. Han, 'Domain-specific hardware accelerators for deep learning,' *Proceedings of the IEEE*, vol. 108, no. 12, pp. 2185-2207, 2020. doi: 10.1109/JPROC.2020.3014798
- [46] N. Moustafa and J. Slay, 'UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set),' in *Proc. IEEE MilCIS*, 2015, pp. 1-6. doi: 10.1109/MilCIS.2015.7348942
- [47] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, 'Toward generating a new intrusion detection dataset and intrusion traffic characterization,' in *Proc. 4th ICISSP*, 2018, pp. 108-116. doi: 10.5220/0006639801080116
- [48] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, 'SMOTE: Synthetic minority over-sampling technique,' *Journal of Artificial Intelligence Research*, vol. 16, pp. 321-357, 2002. doi: 10.1613/jair.953